Systems Biology of Sororin (CDCA5): A Disorder-Centred View of a Fuzzy Cohesin Interaction Hub and its Phospho- and Ubiquitin-Regulated Interactome

Amirhossein Rezaei

Physics of Data, Department of Physics and Astronomy "Galileo Galilei", University of Padova, I-35131 Padova, Italy, amirhossein.rezaeigolmarz@studenti.unipd.it

Abstract

Sororin (CDCA5) stabilises sister-chromatid cohesion by antagonising the cohesin-release factor WAPL, and its activity is switched off during mitosis through phosphorylation. We analysed CDCA5 (UniProt Q96FF9) from a systems-biology perspective by integrating a STRING/curated interaction network, per-residue disorder and binding predictions (IUPred2 and ANCHOR2), UniProt sequence annotations, and the primary literature on its phospho- and ubiquitin-dependent regulation. The CDCA5 network formed a dense cohesin-centred module (16 nodes, 107 edges, density = 0.89) in which Sororin was a high-degree node connected with near-maximal confidence to PDS5A/B, WAPL and the cohesin core, with the cancer-relevant regulators ERK2 (MAPK1) and SPOP appearing only as peripheral, literature-curated edges. Disorder prediction indicated that 61.5% of the 252-residue protein is intrinsically disordered, and that the experimentally mapped interaction determinants—the APC/C KEN box, the CDK1/ERK phospho-sites and the PDS5-binding FGF motif—lie within intrinsically disordered regions, whereas the only folded element is the short C-terminal Sororin domain, which PSIPRED secondary-structure prediction (83% helix/strand) and DISOPRED3 (0% disorder) independently recover as the single ordered region, with DISOPRED3 reproducing the overall disorder fraction (63.9% vs 61.5% by IUPred2). Six of the seven proline-directed (S/T-P) phospho-sites, including the ERK2 target Ser209, lie within these disordered regions, and the seventh in the short linker adjoining the FGF motif; none is in the folded C-terminal domain. To move beyond a generic disorder score and connect the sequence directly to individual network edges, we additionally screened the sequence against the ELM (Eukaryotic Linear Motif) resource: this independently recovered the annotated APC/C KEN box, identified two candidate SPOP-binding degrons at residues 122–126 and 155–159, a MAPK docking motif at residues 77–84 adjacent to the ERK site Ser79, and a PLK1 phosphorylation motif at residues 201–207, so that four of the five network partners scrutinised at motif level (APC/C, SPOP, ERK2/MAPK1, PLK1) are now anchored to a specific short linear motif rather than to disorder alone. A dedicated context-dependent-binding predictor (FuzPred) independently classified most of the protein (63%) as context-dependent and multi-mode (68% of residues with a high multiplicity of binding modes), with only the C-terminal domain folding on binding. Taken together, the results indicate that Sororin operates as a "fuzzy" interaction hub whose partners are engaged through disordered segments and short linear motifs rather than a globular scaffold, providing a structural rationale for its dense, phosphorylation- and ubiquitylation-switchable interactome.

Key words: Sororin; CDCA5; cohesin; intrinsic disorder; fuzzy complex; short linear motif; phospho-interactome; SPOP

1 INTRODUCTION

Faithful segregation of the genome requires that the two sister chromatids produced during DNA replication remain physically tethered until anaphase. This tethering, sister-chromatid cohesion, is provided by the cohesin complex, a ring formed by SMC1A, SMC3 and the kleisin RAD21 together with a STAG1/STAG2 subunit [1]. Cohesin is dynamically loaded and released from chromatin, and the balance between retention and release is set by two antagonistic regulators that both dock onto the HEAT-repeat subunit PDS5: WAPL, which opens the ring and releases cohesin, and Sororin, which locks it [3, 7].

Sororin, encoded by *CDCA5*, was originally identified as a substrate of the anaphase-promoting complex/cyclosome (APC/C) that is required for cohesion in vertebrates [2]. It is a small (252-residue) protein that is expressed and active only after S phase; it counteracts WAPL by competing for a shared binding surface on PDS5, thereby stabilising the acetylated, cohesive state of cohesin [3]. During prophase this protective activity is dismantled: mitotic kinases phosphorylate Sororin, displacing it from PDS5 and unleashing WAPL to remove cohesin from chromosome arms [4, 6]. Beyond the cell cycle, CDCA5 is over-expressed in several carcinomas; an ERK/MAPK phospho-site (Ser209) has been implicated in lung carcinogenesis [8], and more recently CDCA5 was shown to be a substrate of the Cullin-3/SPOP E3 ubiquitin ligase whose degradation modulates prostate-cancer progression [9].

Although the individual partners of Sororin are well catalogued, its interactions have not, to our knowledge, been examined jointly against the structural state of the sequence that mediates them. Sororin has no reported globular fold apart from a short conserved C-terminal element [5], which raises the question of how such a small protein can act as a switchable hub for the cohesin machinery. The central question of this project is therefore whether the partners of Sororin are engaged through folded domains or through intrinsically disordered regions (IDRs) [17], whether the phosphorylation events that toggle these interactions are themselves located in disordered segments, and, more specifically, whether individual network partners can be anchored to a short linear motif (SLiM) rather than to disorder as a general property alone—since it is the motif, not the disorder score, that most directly explains a given network edge. This framing connects the systems-biology description of CDCA5 to the concept of "fuzzy" complexes, in which disordered regions and short linear motifs retain conformational heterogeneity even in the bound state and provide a substrate for regulation by post-translational modification [10, 14, 15].

To translate this question into a computable form, we assembled a CDCA5-centred interaction network from STRING and curated it against the literature, computed per-residue disorder and disordered-binding propensities along the full sequence with IUPred2 and ANCHOR2, mapped the experimentally defined interaction motifs and phospho-sites onto the resulting disorder profile, and screened the sequence against the ELM linear-motif resource to link individual network partners to candidate short motifs. The interaction network then defines *who* binds Sororin, the disorder profile defines *where* on the sequence they bind and in *what* structural context, the linear-motif prediction identifies *which* short element mediates a given edge, and the phospho-site map defines *how* these contacts are switched.

2 METHODS

2.1 Dataset assembly

The reference sequence and functional annotation of human Sororin were retrieved from the UniProt entry Q96FF9 (CDCA5_HUMAN, 252 residues) [27], including curated region, motif and modified-residue features. The CDCA5-centred protein–protein interaction network was obtained from the STRING database (v12, *Homo sapiens*) [24], using the interaction-partners and network endpoints. Two regulatory partners that are established in the primary literature but absent from the high-confidence STRING neighbourhood, ERK2 (MAPK1) [8] and SPOP [9], were added as curated edges to Sororin. Interaction determinants and post-translational modifications used for the mapping analysis (the KEN box, the FGF motif, the CDK1/ERK phospho-sites and the C-terminal Sororin domain) were taken from UniProt feature annotations and cross-checked against the corresponding experimental studies [2, 5–8]. Candidate short linear motifs (SLiMs) were predicted with the ELM (Eukaryotic Linear Motif) resource [23], submitting the Q96FF9 accession to the ELM prediction pipeline.

2.2 Methods used for analysis

Interaction network. STRING interactions with a combined score below 0.4 were discarded and the remaining associations were represented as an undirected graph $G = (V, E)$. Network topology was summarised by the numbers of nodes $|V|$ and edges $|E|$, the density $2|E|/(|V|(|V| - 1))$, and the degree of each node; the confidence of each Sororin edge was recorded as its STRING combined score. As an independent check, the same module was reconstructed and visualised in Cytoscape 3.10 using the stringApp [25, 26], and its topology (degree, betweenness and clustering coefficient) was recomputed with the Cytoscape network analyzer. Functional enrichment (Gene Ontology, KEGG and Reactome pathways) of the 16-protein module was retrieved with the stringApp against the whole-genome background, with terms considered significant at a false-discovery rate below 0.05. Nodes were annotated by functional class (cohesin core, cohesin regulators, mitotic kinases, APC/C–separase, ubiquitin ligase).

Disorder and disordered-binding prediction. Per-residue intrinsic disorder was predicted with IUPred2 in "long" mode, and disordered protein-binding propensity with ANCHOR2, using the reference implementation and energy matrices of the method [21, 22]; residues with IUPred2 > 0.5 were classified as disordered. For every annotated region and interaction motif we computed the mean IUPred2 and ANCHOR2 scores and the fraction of disordered residues, and we identified contiguous disordered (> 0.5) and ANCHOR (> 0.5) segments along the sequence.

Phospho-interactome mapping. Curated phosphorylation sites from UniProt were classified as proline-directed (S/T-P) when the residue immediately C-terminal was a proline, i.e. as candidate substrates of the proline-directed kinases CDK1 and ERK [4, 8]; each site was then located within the disorder profile and annotated with its reported functional consequence.

Linear motif prediction. The ELM prediction pipeline regex-matches the sequence against ~ 350 curated motif classes and applies its own context filters (local secondary structure, sub-cellular topology, taxonomic range) [23]. Of the matches surviving these filters, we retained those whose ELM class corresponds to a partner already present in the network or literature-curated edges of Section 3.1 (APC/C via the KEN-box class, SPOP via the SPOP-binding-consensus class, ERK2/MAPK1 via the MAPK docking-motif class, PLK1 and CDK1 via their respective kinase-motif classes), and computed the mean IUPred2 score and percentage of disordered residues within each retained motif from the disorder profile above. This yields, for a subset of the network, a direct sequence-level candidate for the interaction rather than only a co-occurrence with disorder.

Secondary-structure and second-predictor disorder analysis. Per-residue secondary structure was predicted with PSIPRED 4.0 and intrinsic disorder with DISOPRED3, both run through the PSIPRED Protein Analysis Workbench on the full sequence [19, 20]. PSIPRED assigns each residue to helix (H), strand (E) or coil (C); DISOPRED3 provides a per-residue disorder call and an independent protein-binding-disorder annotation. These serve two purposes: PSIPRED tests whether the folded/disordered partition inferred above is matched by an actual secondary-structure assignment, and DISOPRED3 provides a second, methodologically independent disorder predictor against which the IUPred2 profile can be cross-checked. Per-region helix/strand/coil fractions and disorder fractions were computed over the annotated regions and motifs. Context-dependent binding behaviour was predicted with FuzPred [11, 12], which returns per-residue disorder-to-order (pDO) and disorder-to-disorder probabilities and the multiplicity of binding modes (MBM), and classifies the sequence into fold-on-binding, context-dependent and stays-disordered regions.

2.3 Statistical analysis

Predictions were evaluated at the residue level. Region-level summaries are reported as means over the residues of each region and as the fraction of residues exceeding the disorder threshold; network confidence is reported as STRING combined scores in the interval $[0, 1]$. No inferential test was applied, as the analysis is a descriptive integration of sequence, network and predictor outputs rather than a comparison between samples.

3 RESULTS

3.1 The Sororin interactome is a dense cohesin-centred module

We first asked which proteins define the immediate interaction neighbourhood of Sororin and how tightly they are interconnected. The CDCA5-centred STRING network, restricted to the cohesin machinery and its mitotic regulators, formed a single dense module of 16 nodes and 107 edges with a density of 0.89 (Figure 1), a connectivity comparable to other tightly coupled cell-cycle modules. Sororin itself was a high-degree node (degree $= 15$ of a possible 15), and its edges to the cohesin regulators and core were among the most confident in the whole graph: PDS5A (0.999), WAPL (0.998), SMC3 (0.994), PDS5B

Table 1: *High-confidence STRING partners of Sororin (CDCA5) and their functional class. Combined scores are STRING confidence values in* $[0, 1]$*; the full module comprises 16 nodes and 107 edges (density* $= 0.89$*).*

Partner	Functional class	STRING combined score
PDS5A	Cohesin regulator (PDS5)	0.999
WAPL	Cohesin regulator (release)	0.998
SMC3	Cohesin core	0.994
PDS5B	Cohesin regulator (PDS5)	0.990
RAD21	Cohesin core (kleisin)	0.988
CDK1	Mitotic kinase	0.979
STAG2	Cohesin core	0.977
STAG1	Cohesin core	0.975
SMC1A	Cohesin core	0.973
NIPBL	Cohesin loader	0.970
AURKB	Mitotic kinase	0.968
CDC20	APC/C activator	0.938
ESPL1	Separase	0.930
MAU2	Cohesin loader	0.917
PLK1	Mitotic kinase	0.893
ERK2 (MAPK1)	Cross-talk (literature)	—
SPOP	Ubiquitin ligase (literature)	—

(0.990) and RAD21 (0.988), followed by the mitotic kinase CDK1 (0.979) and the STAG and NIPBL–MAU2 loader subunits (Table 1). The APC/C activator CDC20 and the protease separase (ESPL1) were also recovered, consistent with Sororin being both a cohesion factor and an APC/C substrate.

The two cancer-relevant regulators of the theme behaved differently from the cohesin partners. ERK2 (MAPK1) and SPOP did not appear in the high-confidence STRING neighbourhood of CDCA5 and were included as literature-curated edges (dashed, Figure 1); in the combined graph they remained peripheral, low-degree nodes attached almost exclusively to Sororin. An independent reconstruction of the same module in Cytoscape via the stringApp reproduced the topology exactly (16 nodes, 107 edges, density 0.892, average degree 13.4, network diameter 2, a single connected component and a clustering coefficient of 0.925; Figure 2), with CDCA5 among the maximally connected nodes (degree 15), confirming that the module is a near-complete graph and that the result does not depend on the analysis implementation. This topology indicates that the canonical, high-confidence interactome of Sororin is essentially the cohesin cohesion–release machine, whereas the oncogenic ERK2 and SPOP inputs are regulatory cross-talk nodes that are not yet embedded in the curated cohesin map. Taken together, the network shows that Sororin is a central, high-confidence hub of the cohesin module and simultaneously a target of at least two peripheral regulatory pathways relevant to cancer.

To confirm that this module represents a coherent biological function rather than an arbitrary selection of proteins, we retrieved functional enrichment for the sixteen nodes with the stringApp. The module was far more interconnected than expected for a random set of proteins of the same size (STRING PPI enrichment $p < 10^{-16}$), and its enriched annotations were dominated, at very low false-discovery rates, by precisely the expected biology (Table 2): the leading Gene Ontology biological processes were sister-chromatid cohesion (GO:0007062, FDR $= 1.4 \times 10^{-24}$) and sister-chromatid segregation (FDR $= 2.3 \times 10^{-24}$); the top cellular components were the chromosomal and centromeric regions; the enriched KEGG pathway was cell cycle (hsa04110, FDR $= 6 \times 10^{-14}$); and the leading Reactome pathways were cohesin loading onto chromatin and the establishment and resolution of sister-chromatid cohesion. This enrichment demonstrates quantitatively that the high-confidence CDCA5 neighbourhood analysed here is the sister-chromatid-cohesion machinery itself, providing a functional anchor for the interaction-determinant and disorder analyses that follow.

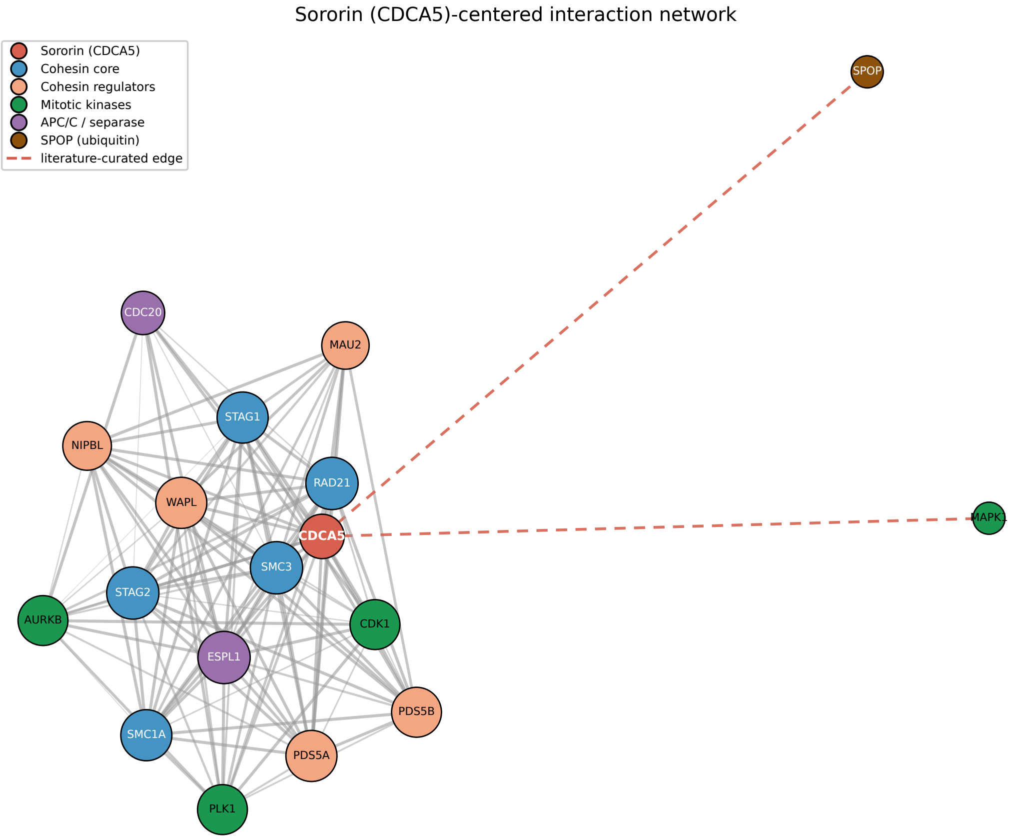

Figure 1: ***Sororin (CDCA5)-centred interaction network.*** *Nodes are proteins coloured by functional class and edges are STRING associations (combined score* ≥ 0.4*); edge width and opacity scale with confidence. Node size scales with degree. Dashed red edges denote literature-curated regulatory interactions (ERK2/MAPK1 and SPOP) that are absent from the high-confidence STRING neighbourhood. The STRING module is a dense cohesin-centred graph (16 nodes, 107 edges, density* $= 0.89$*) with Sororin as a high-degree hub; ERK2/MAPK1 and SPOP are shown as two additional literature-curated nodes.*

3.2 Interaction determinants map to disordered regions: Sororin as a fuzzy hub

Having established the partners, we next asked in what structural context they are bound, i.e. whether the sequence elements that mediate these interactions are folded or intrinsically disordered. Running IUPred2 and ANCHOR2 over the full 252-residue sequence showed that Sororin is predominantly disordered: 61.5% of residues had IUPred2 > 0.5 (mean score 0.61), organised into three long disordered regions (residues 1–48, 72–142 and 199–222) separated by short segments of transient structure (Figure 3). The only region with a robustly folded signature was the C-terminal conserved Sororin domain (residues 230–252; mean IUPred2 0.31, 0% disordered), which is precisely the element shown experimentally to be required for cohesion and cohesin association [5].

Superimposing the experimentally defined interaction determinants onto this profile revealed that they lie, almost without exception, in disordered segments (Table 3). The APC/C-recognition KEN box (residues 88–90) sits in a fully disordered region (mean IUPred2 0.64); the ERK2 phospho-acceptor Ser209 lies on a disorder peak within the 199–222 IDR (IUPred2 0.74); and the PDS5-binding FGF motif (residues 166–168) maps to a short, locally ordered dip embedded within otherwise disordered sequence—the characteristic signature of a short linear motif / molecular-recognition feature that becomes structured only in context, i.e. local order grafted onto a disordered carrier [13, 18]. Consistent with a binding-competent disordered protein, ANCHOR2 predicted several disordered-binding segments spanning the N-terminal, central and C-terminal IDRs (residues 1–70, 78–118, 127–153, 198–204 and 227–240), which collectively cover the KEN box, the phospho-cluster and the region approaching the FGF motif. The contrast between the folded C-terminal domain and the disordered location of the KEN box, the FGF motif and the phospho-sites in-

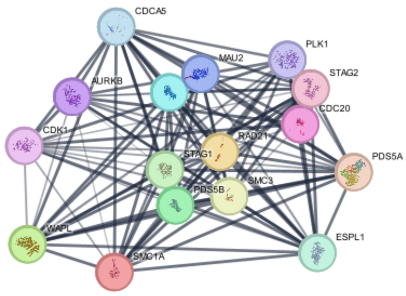

Figure 2: ***Independent reconstruction of the cohesin module in Cytoscape.*** *The same 16-protein module was retrieved from STRING and rendered with the Cytoscape stringApp; node glyphs show STRING structure previews and edge thickness scales with the combined score. Cytoscape's network analyzer returned the same topology as the custom pipeline (16 nodes, 107 edges, density* 0.892*, clustering coefficient* 0.925*), with CDCA5 a maximally connected node (degree* 15*).*

Table 2: *Representative functional enrichment of the 16-protein module (stringApp, whole-genome background, FDR* < 0.05*). The module shows a STRING PPI enrichment* $p < 10^{-16}$*; the terms below are the most significant in each source, all reflecting sister-chromatid cohesion and cell-cycle function.*

Source	Term	Description	FDR
GO Process	GO:0007062	Sister chromatid cohesion	1.4×10^{-24}
GO Process	GO:0000819	Sister chromatid segregation	2.3×10^{-24}
GO Process	GO:0000278	Mitotic cell cycle	1.2×10^{-18}
GO Component	GO:0098687	Chromosomal region	7.0×10^{-15}
KEGG	hsa04110	Cell cycle	6.1×10^{-14}
Reactome	HSA-2500257	Resolution of sister chromatid cohesion	1.7×10^{-23}
Reactome	HSA-2468052	Establishment of sister chromatid cohesion	4.4×10^{-22}

dicates that Sororin engages its partners chiefly through disordered segments and short linear motifs rather than through a globular interface. Taken together, these results support the view that Sororin behaves as a fuzzy interaction hub, in which a largely disordered chain presents multiple recognition elements whose local structure and accessibility can be tuned by modification.

We tested this fuzzy-hub interpretation directly with FuzPred, a sequence-based predictor of context-dependent binding behaviour built specifically to quantify fuzziness [11, 12]. Consistent with a fuzzy hub, Sororin was predicted to engage partners predominantly without folding: the median disorder-to-order probability was only 0.40, 62% of residues favoured disorder-to-disorder (fuzzy) binding, and 68% showed a high multiplicity of binding modes (MBM > 0.65), i.e. the capacity to bind in several alternative modes (Figure 4). FuzPred classified 63% of the sequence as context-dependent and only 25% as fold-on-binding, the latter confined to three short segments including the C-terminal Sororin domain (residues 231–248). Mapping the interaction determinants onto this landscape reproduced the division of labour identified above: the C-terminal domain is fold-on-binding (pDO 0.66), whereas the PDS5-binding FGF motif (pDO 0.34, pDD 0.66, MBM 0.70) and the ERK2 site Ser209 (pDO 0.33, MBM 0.80) lie in fuzzy, multi-mode regions, while the APC/C KEN box is the expected exception, predicted to fold upon binding (pDO 0.87) as befits a degron recognised by a folded receptor. FuzPred additionally annotated the two Pfam families of Sororin—the middle region (89–214) and the C-terminal region (229–252)—of which only the C-terminal family coincides with the ordered, fold-on-binding segment. That an independent method developed specifically to detect fuzziness returns this multi-mode, mostly-disordered binding picture provides quantitative, method-orthogonal support for describing Sororin as a fuzzy interaction hub.

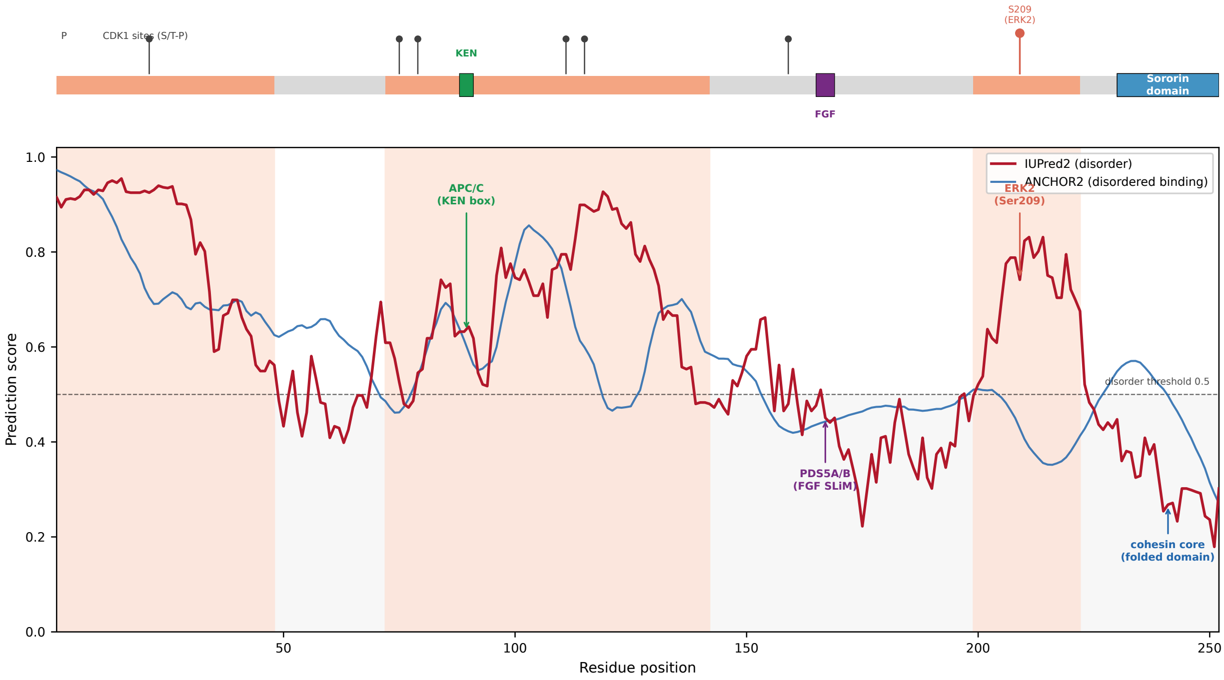

Figure 3: ***Disorder landscape of Sororin with interaction determinants annotated.*** *Top, domain architecture: three intrinsically disordered regions (orange; UniProt 1–48, 72–142, 199–222), the folded C-terminal Sororin domain (blue), the KEN box and FGF motif, and proline-directed (S/T-P) phospho-sites (lollipops; the ERK2 target Ser209 in red). Bottom, per-residue IUPred2 disorder (red) and ANCHOR2 disordered-binding (blue) scores; the dashed line is the disorder threshold* (0.5) *and shaded bands mark the annotated IDRs. The APC/C KEN box and the ERK2 site Ser209 fall in disordered regions, the PDS5-binding FGF motif is a short ordered dip embedded in disorder, and the only folded element is the C-terminal Sororin domain.*

3.3 The phospho-interactome and SPOP ubiquitin cross-talk act through the disordered chain

Finally, we asked how the disordered interactome of Sororin is switched, by locating its regulatory modifications on the same profile. Of the thirteen curated phospho-sites, seven are proline-directed (S/T-P) and therefore candidate substrates of the proline-directed kinases CDK1 and ERK: Ser21, Ser75, Ser79, Thr111, Thr115, Thr159 and Ser209 (Table 4). None lies in the folded C-terminal domain: six fall within the three intrinsically disordered regions and the seventh, Thr159, in the short linker immediately preceding the FGF motif. They flank the very determinants identified above: the central cluster (Ser75–Thr115) surrounds the KEN box, Thr159 abuts the PDS5-binding FGF motif, and Ser209 sits in the C-terminal IDR. This co-localisation provides a structural explanation for the established regulatory logic in which mitotic phosphorylation of Sororin by CDK1 (assisted by Aurora B) displaces it from PDS5 and relieves WAPL inhibition, dissolving arm cohesion [4, 6]: because the PDS5-binding FGF motif is a short linear element embedded in a disordered, heavily phosphorylatable context, phosphorylation of the surrounding chain can switch the interaction off without requiring the disruption of a folded domain, exactly as expected for a fuzzy complex [10, 14].

The same disordered architecture rationalises the two cancer-associated inputs. The ERK2/MAPK site Ser209, reported to be phosphorylated during lung carcinogenesis [8], is one of the most disordered positions in the protein, i.e. fully accessible to a proline-directed kinase; and the SPOP-dependent degradation that controls CDCA5 levels in prostate cancer [9] requires a SPOP-binding degron, a class of short linear motif that operates only in disordered, accessible sequence. Thus both the phospho-switch that governs cohesion and the ubiquitin-dependent control of Sororin abundance are implemented on the same disordered scaffold. Taken together, the phospho-site map indicates that the regulation of Sororin—mitotic inactivation, oncogenic ERK phosphorylation and SPOP-mediated turnover—is concentrated on its intrinsically disordered regions, consistent with a fuzzy hub whose interactions are designed to be toggled by post-translational modification.

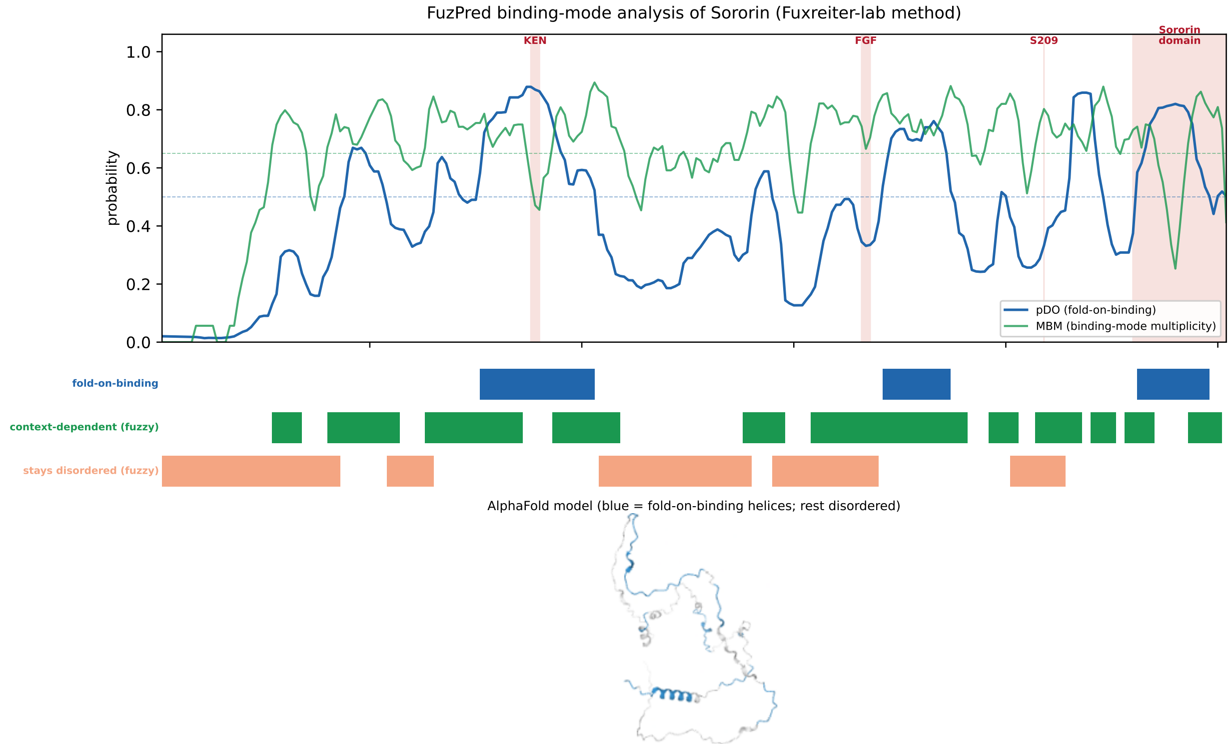

Figure 4: ***FuzPred binding-mode analysis of Sororin.*** *Top, per-residue probability of disorder-to-order (fold-on-binding) binding (pDO, blue) and multiplicity of binding modes (MBM, green); dashed lines mark the* 0.5 *and* 0.65 *references and red bands the interaction determinants. Middle, FuzPred region classification in three overlapping tracks (fold-on-binding, context-dependent, and stays-disordered). Bottom, the AlphaFold model (fold-on-binding helices in blue, the remainder disordered). Most of the protein is context-dependent and multi-mode; only three short segments, including the C-terminal domain, fold on binding.*

3.4 Predicted short linear motifs anchor specific network edges to the disordered chain

The disorder profile shows that Sororin's interaction determinants sit in disordered sequence, but disorder alone does not specify which partner binds where. To close this gap we screened the full sequence against the ELM resource and retained the predictions whose motif class corresponds to a partner already present in the network (Table 5, Figure 5). ELM independently recovered the annotated KEN box (residues 87–91, flagged by ELM as an experimentally validated instance) as an APC/C-recognition (DEG_APCC_KENBOX_2) motif, matching the UniProt annotation (88–90) to within one residue. It further identified two SPOP-binding-consensus degrons (DEG_SPOP_SBC_1) at residues 122–126 and 155–159, providing a concrete sequence-level candidate for the SPOP-dependent degradation reported in prostate cancer [9], which the network alone could only represent as an edge. A generic MAPK docking motif (DOC_MAPK_gen_1, residues 77–84) was found immediately adjacent to the reported ERK/MAPK phospho-site Ser79 [8], giving ERK2 a plausible physical docking element in addition to its phospho-acceptor site, and a PLK1 phosphorylation motif (MOD_Plk_1) was identified at residues 201–207, close to the ERK2 site Ser209, consistent with PLK1's role as one of the mitotic kinases in the high-confidence STRING module. Finally, nine occurrences of the generic proline-directed kinase motif (MOD_ProDKin_1) recapitulated the same cluster of CDK1 candidate sites as the manual S/T-P classification of Section 3.3, here obtained through ELM's curated motif pattern rather than a hand-coded rule, providing a consistency check between the two approaches.

Of these fourteen network-linked motifs, eleven (79%) are majority disordered (mean IUPred2 0.65, mean 67% disordered over the motif window for the CDK1 cluster); the exceptions are the second SPOP degron and one CDK1 site, which fall in the short, partially ordered linker adjoining the FGF motif, in the same region as the outlier phospho-site Thr159 identified in Section 3.3. Thus, in contrast to reporting disorder as a general property of the protein, this analysis assigns a specific short linear motif to four of the

Table 3: *Mapping of Sororin interaction determinants onto the predicted disorder landscape. Region disorder is the mean IUPred2 score and the fraction of residues with IUPred2 $>$ 0.5 over the indicated residues.*

Determinant	Residues	Partner / role	Mean IUPred2 (% dis.)	Structural context
N-terminal IDR	1–48	loading / regulation	0.82 (100%)	disordered
KEN box	88–90	APC/C (degradation)	0.64 (100%)	disordered
Central IDR	72–142	phospho-cluster	0.70 (90%)	disordered
FGF motif	166–168	PDS5A/B (cohesion)	0.47 (33%)	SLiM / ordered dip in IDR
Ser209 site	199–222	ERK2 (phosphorylation)	0.71 (96%)	disordered
Sororin domain	230–252	cohesin core association	0.31 (0%)	folded

Table 4: *Proline-directed (S/T-P) phosphorylation sites of Sororin and their context. Six of the seven candidate CDK1/ERK sites lie within intrinsically disordered regions and the seventh (Thr159) in the short linker preceding the FGF motif; none is in the folded C-terminal domain.*

Site	Motif (S/T-P)	Disordered region	Reported role / kinase
Ser21	SP	1–48 (N-terminal IDR)	CDK1 consensus [4]
Ser75	SP	72–142 (central IDR)	CDK1 consensus [4]
Ser79	SP	72–142 (central IDR)	ERK/MAPK site [8]
Thr111	TP	72–142 (central IDR)	CDK1 consensus [4]
Thr115	TP	72–142 (central IDR)	CDK1 consensus [4]
Thr159	TP	linker (pre-FGF)	CDK1 consensus [4]
Ser209	SP	199–222 (C-terminal IDR)	ERK2 site, lung cancer [8]

five scrutinised network partners (APC/C, SPOP, ERK2/MAPK1 and PLK1), each located predominantly, though not exclusively, within a disordered region.

3.5 Secondary-structure prediction and a second disorder predictor recover the single folded element

Finally, we asked whether the folded/disordered partition inferred from IUPred2 is matched by an actual secondary-structure assignment and by an independent disorder predictor. PSIPRED 4.0 assigned only 18.7% of the sequence to helix and 3.2% to strand, leaving 78.2% as coil (Figure 6), and placed its only two extended helices at residues 133–145 and, most prominently, 225–245; the latter lies within the conserved C-terminal Sororin domain (residues 230–252), which was 70% helix and 83% ordered secondary structure overall (Table 6). DISOPRED3 independently classified 63.9% of the protein as disordered, closely matching the IUPred2 value of 61.5% obtained by a different method (Section 3.2), and assigned the C-terminal Sororin domain 0% disorder. The two predictors therefore converge on the same architecture: the C-terminal domain is the single element that is simultaneously ordered in secondary structure (83% H/E) and non-disordered (0%), whereas the three intrinsically disordered regions are almost entirely coil (79–100%) and predominantly disordered (46–100% by DISOPRED3). The interaction motifs mapped, as before, outside the folded domain: the KEN box, the MAPK docking motif and both SPOP degrons fell in coil, and the PDS5-binding FGF motif was assigned coil with a locally low DISOPRED3 disorder score, consistent with the locally ordered but non-regular SLiM dip identified in Section 3.2 and with the principle that a linear motif is local order grafted onto a disordered carrier [13]. Taken together, an actual secondary-structure prediction and a second, independent disorder predictor both recover the one experimentally supported folded element [5] and assign no comparable ordered structure to the disordered interaction regions, corroborating the disorder-centred architecture with methods distinct from the IUPred2/ANCHOR2 analysis.

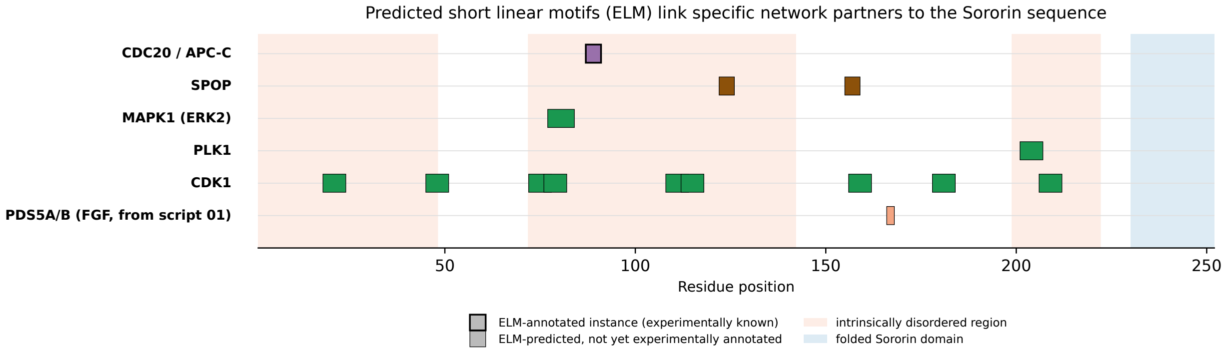

Figure 5: *Predicted short linear motifs link specific network partners to the Sororin sequence.* *Each row is a network partner (or literature-curated node); coloured blocks mark the residue range of an ELM-predicted motif for that partner, coloured as in Figure 1. Thick-edged blocks are ELM-annotated (experimentally documented) instances; thin-edged blocks are ELM predictions without a prior experimental annotation. Shaded bands as in Figure 3: orange for the intrinsically disordered regions, blue for the folded C-terminal domain. The FGF motif (UniProt annotation, not an ELM class) is shown for reference.*

Table 5: *Curated, network-linked short linear motifs predicted by ELM. The CDK1 row summarises the nine MOD_ProDKin_1 hits (individual ranges: 18–24, 45–51, 72–78, 76–82, 108–114, 112–118, 156–162, 178–184, 206–212); all other rows are single motifs. Disorder statistics are the mean IUPred2 score and % of residues with IUPred2* > 0.5 *over the motif window(s), from the same profile as Table 3.*

ELM class	Residues	Partner	Mean IUPred2	% dis.	Annotated?
DEG_APCC_KENBOX_2	87–91	CDC20 / APC-C	0.63	100%	yes (ELM)
DEG_SPOP_SBC_1	122–126	SPOP	0.85	100%	no
DEG_SPOP_SBC_1	155–159	SPOP	0.51	40%	no
DOC_MAPK_gen_1	77–84	MAPK1 (ERK2)	0.59	75%	no
MOD_Plk_1	201–207	PLK1	0.67	100%	no
MOD_ProDKin_1 (9 sites)	see caption	CDK1	0.65	67%	no

4 DISCUSSION AND CONCLUSION

We set out to determine whether Sororin engages its partners through folded domains or through intrinsically disordered regions, and whether the modifications that toggle those interactions are located in disordered sequence. Integrating a cohesin-centred interaction network with per-residue disorder and binding predictions and with the experimentally defined interaction motifs, we found that Sororin is a small, high-confidence hub of a dense cohesin module (density 0.89) that is itself $\sim$62% intrinsically disordered; that its APC/C, PDS5 and ERK2 determinants all reside in disordered segments while only the short C-terminal domain is folded; and that its seven proline-directed phospho-sites lie outside the folded C-terminal domain, in the disordered regions and adjoining linker surrounding those determinants; and, going beyond disorder as a general property, that an independent linear-motif prediction anchors four of the five scrutinised network partners (APC/C, SPOP, ERK2/MAPK1, PLK1) to a specific short sequence element, predominantly but not exclusively within the disordered regions. These observations answer the question directly: the Sororin interactome is built on disorder, and where a partner-specific claim can be made at all, it is a short linear motif embedded in that disorder, rather than disorder in the abstract, that explains the edge.

This picture is consistent with, and adds a structural layer to, the established biology of Sororin. Nishiyama and colleagues demonstrated that mitotic phosphorylation of Sororin by Cdk1 and Aurora B removes it from PDS5 and activates WAPL [6], and Dreier *et al.* showed that Cdk1 phosphorylation regulates Sororin [4]; our finding that the proline-directed sites cluster in the disordered regions flanking the PDS5-binding motif supports a mechanism in which the switch is implemented on a disordered, modification-dense chain rather than through a rigid interface. The crystal structure of PDS5B bound to the shared FGF peptide of WAPL and Sororin [7] shows that this contact is made by a short linear motif presented from otherwise unstruc-

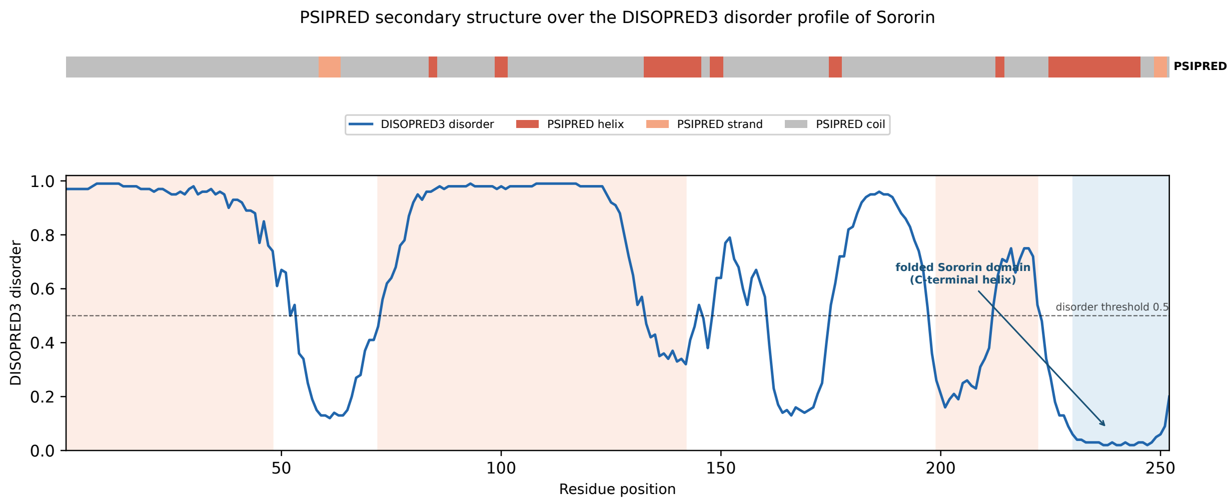

Figure 6: ***PSIPRED secondary structure over the DISOPRED3 disorder profile of Sororin.*** *Top, per-residue PSIPRED assignment (red, helix; orange, strand; grey, coil). Bottom, DISOPRED3 per-residue disorder with the* 0.5 *threshold (dashed); shaded bands as in Figure 3: orange, the three intrinsically disordered regions; blue, the folded C-terminal Sororin domain. The only extended ordered element is the C-terminal helix (residues 225–245), which coincides with the deep DISOPRED3 disorder minimum; the remainder of the chain is coil and predicted disordered.*

Table 6: *PSIPRED secondary structure and DISOPRED3 disorder over the annotated regions and network-linked motifs. Values are the percentage of residues in each region assigned to helix, strand or coil by PSIPRED, and to the disordered state by DISOPRED3. Only the C-terminal Sororin domain is ordered by both measures.*

Region	Residues	% helix	% strand	% coil	% disordered
N-terminal IDR	1–48	0	0	100	100
Central IDR	72–142	21	0	79	85
C-terminal IDR	199–222	8	0	92	46
KEN box (APC/C)	88–90	0	0	100	100
MAPK docking (ERK2)	77–84	12	0	88	100
SPOP degron 1	122–126	0	0	100	100
FGF motif (PDS5A/B)	166–168	0	0	100	0
PLK1 motif	201–207	0	0	100	0
Sororin domain	230–252	70	13	17	0

tured sequence, in agreement with our observation that the FGF motif is a locally ordered dip embedded in disorder—the local order within a disordered carrier that is the defining structural signature of linear motifs [13]. In addition, the requirement of the conserved C-terminal Sororin element for cohesion [5] matches the single folded region in our profile—recovered independently by both disorder and secondary-structure prediction as the one ordered, helical element—suggesting a division of labour between a disordered, regulatory body and a compact functional C-terminus. Our results therefore align with, and give insight into, prior experimental work by explaining *why* Sororin's interactions are so readily switchable.

The framework also accommodates the two cancer-associated regulators that were only peripheral in the network. The ERK2 site Ser209, phosphorylated in lung carcinogenesis [8], is among the most disordered and accessible residues of the protein and lies close to a predicted PLK1 phosphorylation motif and, further upstream, to a predicted MAPK docking motif adjoining the second reported ERK site Ser79; and the SPOP-dependent degradation that controls CDCA5 abundance and prognosis in prostate cancer [9] depends on a degron that, like all SPOP substrate motifs, must be presented from accessible, disordered sequence—consistent with the two SPOP-binding-consensus motifs predicted here at residues 122–126 and 155–159. Placing these inputs on the disorder map suggests a unifying model: Sororin is a fuzzy interaction hub [10, 14]—a description independently supported by FuzPred, which classifies the majority of the chain as context-dependent, multi-mode binding [11, 12]—consistent with the enrichment of intrinsic disorder among interaction hubs [16], in which a mostly disordered chain displays a set of short linear motifs—the KEN degron, the FGF motif, two candidate SPOP degrons and a MAPK docking motif—interleaved with proline-directed phospho-sites, so that kinases and ubiquitin ligases can turn individual contacts on or off without remodelling a folded core. We propose that the very disorder of Sororin is what makes it an efficient, multiply-regulated node: it converts a small protein into a programmable switchboard for cohesin, and that this switchboard is implemented through a small, enumerable set of linear motifs rather than through disorder alone.

Three limitations temper these conclusions. First, disorder and binding propensities are predictions; although IUPred2 and ANCHOR2 are well validated and the folded C-terminal domain and the FGF dip are recovered as expected, direct structural confirmation of the bound states would strengthen the fuzzy-hub interpretation. Second, the ERK2 and SPOP edges rest on individual studies and are not yet part of curated interaction databases, so their network context is provisional. Third, the ELM motif predictions reported here are, with the exception of the KEN box, sequence-based predictions without direct experimental validation in Sororin itself; the SPOP degrons and MAPK docking motif in particular should be considered testable hypotheses—for example by mutagenesis of the SPOP-binding-consensus residues and assessment of CDCA5 turnover—rather than established interaction sites.

In conclusion, a disorder-centred systems analysis indicates that Sororin functions as a fuzzy cohesin hub whose partners—the cohesin core, PDS5A/B and WAPL, the APC/C, and the cancer-linked ERK2 and SPOP pathways—are engaged and switched through intrinsically disordered regions and short linear motifs rather than a globular scaffold; independent ELM motif prediction anchors four of these five partners to a specific, enumerable sequence element rather than to disorder as a general property, providing candidate SPOP degron and MAPK docking sites that the network alone could not localise. Looking forward, the model makes testable predictions: phospho-mimetic substitutions within the disordered central and C-terminal regions should quantitatively tune PDS5 binding and Sororin turnover, and mutation of the predicted SPOP-binding-consensus residues (122–126 or 155–159) should measurably alter CDCA5 turnover if either is the operative degron. Experimental dissection of these disordered elements and their motifs, ideally combined with an ensemble structural description of the Sororin–PDS5 complex, would test whether targeting a disordered hub offers a route to modulating cohesion in cancers where CDCA5 is over-expressed.

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